

WASP-12b

R-0561

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_170205 · created 2026-07-23T17:29:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457788.69377 +/- 0.00036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.045
Transit depth (Rp/Rs)^2	3.9e-05 +/- 0.0056 [%]
Orbital Inclination (inc)	86.46 +/- 2.9
Airmass coefficient 1 (a1)	21.55 +/- 13.31
Airmass coefficient 2 (a2)	-2.89 +/- 1.38
Scatter in the residuals of the lightcurve fit is	58.69 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.0
Transit Duration (day)	0.1173 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.045 vs 0.1178 (deviation 2.6σ)
Duration fit vs published	0.1173 d vs 0.125 d (deviation 1.6σ)
Mid-time O–C	-1571.3 min
Depth SNR	0.0
Residual scatter	58.69 %

Light curve

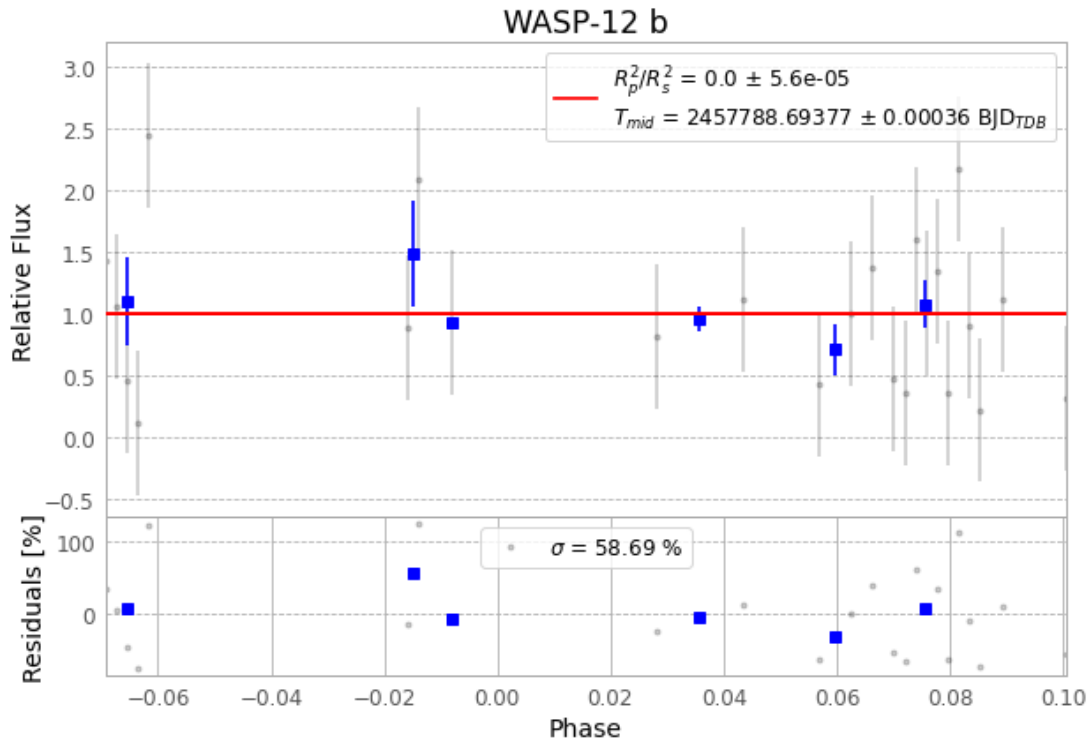


Figure 1 — FinalLightCurve_WASP-12 b_2017-02-03.png (SHA-256 1abb875faf4afdc8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 344], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf35fb838ef7af94...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

93 FITS frames (61 MB), WASP-12170204023912.FITS → WASP-12170204071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170205.log (2c4dbf0d18c5...), FinalLightCurve_WASP-12 b_2017-02-03.png (1abb875faf4a...), FinalLightCurve_WASP-12 b_2017-02-03.csv (5bd39ebceb99...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 17:29Z.